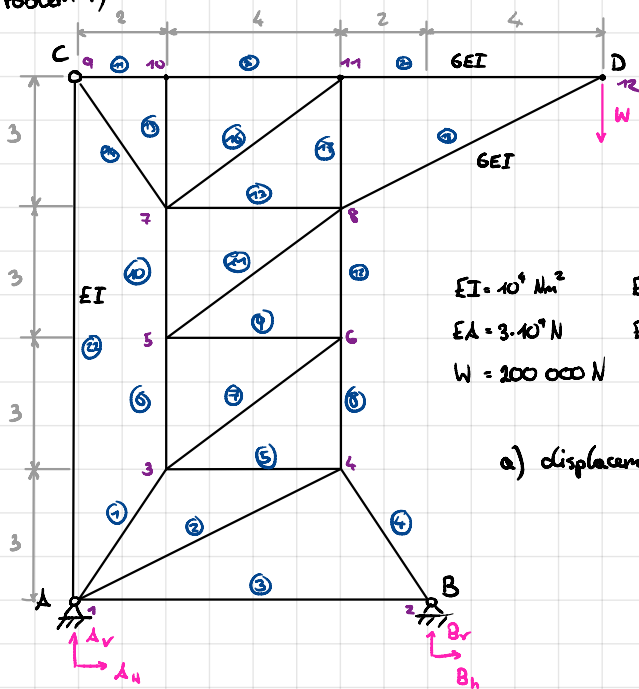


Problem 1)



$$EI = 10^4 \text{ Nm}^2 \quad EI = 10^4 \Rightarrow E = 10^{10} \text{ I} = 0,000001 \text{ m}^4$$

$$EA = 3 \cdot 10^9 \text{ N} \quad EA = 3 \cdot 10^9 \Rightarrow A = \frac{3 \cdot 10^9}{10^{10}} = 0,3 \text{ m}^2$$

$$W = 100\,000 \text{ N}$$

a) displacement in z at 12: $r[11 \cdot 3 + 1] = -0,010284 \text{ m}$

JUPYTER_EXPORT_THOENI_EX5.html

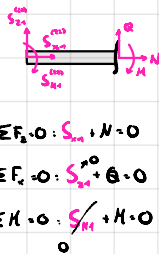
contains a pre-run version of the source code to view without python

b) $\sum F_x: A_H + B_H \stackrel{!}{=} 0 \quad = -2,88 \text{ e-} 9$

$\sum F_y: A_V + B_V + W \stackrel{!}{=} 0 \quad = 3,49 \text{ e-} 10$

c) $\vec{S} = k \vec{r}_c \quad \vec{r}_c = \vec{T}_3 \vec{r}$

$$S^{(m)} = k^{(m)} \begin{bmatrix} r_{10} \cdot 3 \cdot 1 \cdot 3^T \\ r_{18} \cdot 3 \cdot 9 \cdot 3^T \end{bmatrix}$$



$$\sum F_x = 0: S_{i2} + H = 0$$

$$\sum F_y = 0: S_{i1} + Q = 0$$

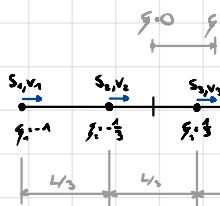
$$\sum M = 0: S_{j1} \cdot L + H \cdot 0 = 0$$

$$N = -S_{i1} = 2,09347 \text{ e} 5 \text{ N}$$

$$Q = 0$$

$$H = 0$$

Problem 2)



$$N_i(x) = \ell_i^n(x) = \prod_{j=1}^n \frac{x - x_j}{x_i - x_j} \quad x = \frac{L}{2}(\xi + 1) \Rightarrow \xi = \frac{2x}{L} - 1$$

$$n=3: N_i(x) = \ell_i^3(x) = \frac{x - x_2}{x_i - x_2} \cdot \frac{x - x_3}{x_i - x_3} \cdot \frac{x - x_4}{x_i - x_4}$$

$$= \ell_i^3(\xi) = \frac{\xi - \xi_2}{\xi_i - \xi_2} \cdot \frac{\xi - \xi_3}{\xi_i - \xi_3} \cdot \frac{\xi - \xi_4}{\xi_i - \xi_4}$$

$$N_1(\xi) = \frac{\xi - \xi_2}{\xi_1 - \xi_2} \cdot \frac{\xi - \xi_3}{\xi_1 - \xi_3} \cdot \frac{\xi - \xi_4}{\xi_1 - \xi_4} = \frac{\xi + \frac{1}{3}}{-\frac{2}{3}} \cdot \frac{\xi - \frac{1}{3}}{-\frac{4}{3}} \cdot \frac{\xi - 1}{-2} = \frac{(\xi + \frac{1}{3})(\xi - \frac{1}{3})(\xi - 1)}{-\frac{16}{9}} = \frac{-9}{16} (\xi + \frac{1}{3})(\xi - \frac{1}{3})(\xi - 1) = N_1(\xi)$$

$$N_2(\xi) = \frac{\xi - \xi_1}{\xi_2 - \xi_1} \cdot \frac{\xi - \xi_3}{\xi_2 - \xi_3} \cdot \frac{\xi - \xi_4}{\xi_2 - \xi_4} = \frac{\xi + 1}{\frac{2}{3}} \cdot \frac{\xi - \frac{1}{3}}{-\frac{2}{3}} \cdot \frac{\xi - 1}{-\frac{4}{3}} = \frac{27}{16} (\xi + 1)(\xi - \frac{1}{3})(\xi - 1) = N_2(\xi)$$

$$N_3(\xi) = \frac{\xi - \xi_1}{\xi_3 - \xi_1} \cdot \frac{\xi - \xi_2}{\xi_3 - \xi_2} \cdot \frac{\xi - \xi_4}{\xi_3 - \xi_4} = \frac{\xi + 1}{\frac{4}{3}} \cdot \frac{\xi + \frac{1}{3}}{\frac{2}{3}} \cdot \frac{\xi - 1}{-\frac{2}{3}} = -\frac{27}{16} (\xi + 1)(\xi + \frac{1}{3})(\xi - 1) = N_3(\xi)$$

$$N_4(\xi) = \frac{\xi - \xi_1}{\xi_4 - \xi_1} \cdot \frac{\xi - \xi_2}{\xi_4 - \xi_2} \cdot \frac{\xi - \xi_3}{\xi_4 - \xi_3} = \frac{\xi + 1}{2} \cdot \frac{\xi + \frac{1}{3}}{\frac{4}{3}} \cdot \frac{\xi - \frac{1}{3}}{\frac{2}{3}} = \frac{9}{16} (\xi + 1)(\xi + \frac{1}{3})(\xi - \frac{1}{3}) = N_4(\xi)$$